

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

1

Use a Punnett square to predict the offspring ratio from two heterozygous parents.

[4 marks]

2

Describe how the body maintains a constant internal temperature.

[5 marks]

3

Explain the role of the placenta in exchanging substances between mother and fetus.

[4 marks]

4

Describe the structure of a cell membrane and explain how it controls what enters the cell.

[6 marks]

5

Explain the difference between diffusion, osmosis, and active transport, giving one example of each.

[6 marks]

6

Describe the role of enzymes and explain why activity falls sharply above an optimum temperature.

[5 marks]

7

Explain how the structure of an alveolus is adapted for gas exchange.

[5 marks]

8

Outline the process of photosynthesis and state the word equation.

[5 marks]
